

Pamella Carneiro

Physical chemist, from the **lab bench** to the **simulation box**.
Den Haag, The Netherlands

Email · LinkedIn · GitHub · ORCID · X / Twitter

I led a water and wastewater analysis team in Belém, Pará, in the Amazon region of Brazil, overseeing laboratory operations, ISO 9001 compliance, and the transition toward ISO/IEC 17025. I coordinated laboratory and field workflows, conducted proficiency testing, and performed physicochemical and microbiological analyses.

My background combines hands-on laboratory work with computational chemistry and data analysis. I have used analytical equipment and computational tools to improve workflows and analyze physicochemical data, with research experience in molecular dynamics, biomolecular systems, solvation, and protein–solvent interactions.

This combination of experimental and computational backgrounds allows me to approach technical and analytical challenges from both laboratory and data-driven perspectives.

	EXPERIMENTAL	COMPUTATIONAL
INDUSTRY	**Physical-chemical analyses** Water & wastewater testing, microbiology, QC under ISO/IEC 17025 and ISO 9001	**Lab informatics** LIMS for metrological traceability and results control · Windows & Linux
RESEARCH	**Experimental Chemistry** Teaching assistant in laboratory analyses	**Molecular modelling** Structure–activity studies, molecular dynamics, antigen–antibody interactions, solvation, and modelling of protein structure and disorder

Common ground: Analytical · Research · Data Analysis · Problem-Solving · Reproducibility.

3 yrs
technical responsibility in a physicochemical lab

2015
doing research since my first scientific initiation

2
peer-reviewed papers as first author

BR → NL
UFPA · USP · UNICAMP · TU Delft

01 Leadership

Taking responsibility for results.

Apr 2018 – Apr 2021

Quality Control `industry`

LAS Sanches Laboratório · Brazil

- Technical responsibility for water and industrial wastewater testing and all laboratory activities
- Evaluated analytical results and test conformity before they reached clients
- Managed quality procedures, analytical records and metrological traceability

Open source

Knowledge sharing `research`

- Teaching assistant guiding students through experimental analyses
- Molecular dynamics tutorial so newcomers can set up and run their first simulations · github.com/pamellaccar
- Teaching assistant in quantum chemistry for the undergraduate course
- Connecting with the scientific community · x.com/pamellachem

2016

Recognition `research`

UFPA · EACH-USP

- Highlight in Scientific and Technological Initiation, Federal University of Pará (UFPA)
- Scientific Initiation Summer Scholarship, Artificial Intelligence Laboratory, EACH - University of São Paulo (USP)
- 3rd Place – Best Poster Award, Encontro dos Profissionais de Química da Amazônia (Meeting of Chemistry Professionals of the Amazon)

02 Physical-chemical analyses

Hands-on bench work on water and industrial wastewater.

Water quality parameters

Routine characterisation of water and effluents.

`pH · conductivity · turbidity · solids · COD/BOD`

Titrimetry & classical methods

`titration · alkalinity · hardness · chlorine`

Instrumental analysis

spectrophotometry · metals · Gallery automated analyzer

Microbiological testing

Sanitary quality of water samples.

total coliforms · Escherichia coli

Where it meets the computer: I used LIMS (Laboratory Information Management System) for result documentation and traceability.

03 Quality control

Apr 2018 – Apr 2021

Laboratory Analyst & Technical Responsibility industry

LAS Sanches Laboratório · Brazil

- Experience with ISO 9001
- Applied ABNT NBR ISO/IEC 17025 requirements to day-to-day laboratory work
- Kept metrological traceability and analytical records
- Followed sampling protocols and assessed conformity of test results
- Conducted internal audits
- Performed proficiency tests

Apr 2018 – Apr 2021

Complying with regulatory standards industry

LAS Sanches Laboratório · Brazil

- Groundwater – CONAMA No. 396/2008
- Drinking Water (Human Consumption) – Consolidation Ordinance No. 5 (Ministry of Health)
- Swimming Pool, Recreational and Primary Contact Water – NBR 10818:1989
- Surface Water – CONAMA No. 357/2005
- Bathing Water Quality (Recreational Water) Analysis – CONAMA No. 274/2000
- Liquid and Sanitary Effluent Analysis – CONAMA No. 430/2011

ABNT NBR ISO/IEC 17025 · Metrological traceability · Conformity assessment · Sampling protocols · Analytical records · Laboratory documentation · Quality procedures

From QC to research: I bring a quality-driven mindset to computational research, applying the same principles of control, traceability, and validation to build reliable simulation workflows and ensure confidence in the results.

04 Drug discovery & medicinal chemistry

Molecular understanding for vaccines and therapeutics: antigen interactions, antibody recognition, and stability and affinity in formulation.

01 · ANTIGEN	02 · RECOGNITION	03 · FORMULATION	04 · QUALITY
Structure & intrinsic disorder	**Antigen–antibody mechanisms**	**Solvation, ionic liquids, adjuvants**	**Analytics & conformity**
comp	comp	comp	exp

comp computational · exp experimental

Antigen–antibody recognition

Enhanced-sampling MD of the malaria antigen PfAMA1 bound to a monoclonal antibody, showing an extended conformational selection mechanism (MSc, J. Phys. Chem. B 2024).

HREMD · GROMACS · clustering

Intrinsically disordered antigens

Review of disordered malaria antigens and how molecular simulation can open them up for drug and vaccine development (J. Braz. Chem. Soc. 2024).

IDPs · force fields · enhanced sampling

Vaccine adjuvants & ionic liquids

Molecular Dynamics to understand how ionic liquids solvate and stabilise antigen proteins, as candidates for vaccine adjuvants.

Kirkwood-Buff theory · MDDF · solvation

Medicinal chemistry roots

Structure–activity studies of bioactive molecules during my scientific initiation at the Molecular Modeling Laboratory (UFPA): 3D-QSAR, HQSAR and docking.

molecular modeling · medicinal chemistry

Why both sides matter: meaningful computational insights depend on reliable experiments, stable formulations and traceable analytical results. I bring experience from both ends.

05 Research

Computational chemistry of biomolecular systems: explaining macroscopic

behaviour from molecular interactions.

Academic path

Mar 2026 – present

Guest Researcher `computational`

Delft University of Technology (TU Delft) · Delft, Netherlands

- Molecular dynamics and computational analysis of molecular and biomolecular systems
- Calculation of transport properties of industrially relevant solvents
- Self and collective diffusivities and viscosity from equilibrium MD (Maxwell–Stefan, Green–Kubo)

Feb 2024 – present

PhD candidate in Physical Chemistry `computational`

University of Campinas (UNICAMP) · Brazil

- Kirkwood–Buff theory and minimum-distance distribution functions to quantify how proteins are solvated in mixed solvents
- Solvation and biomolecular interactions in complex systems: ionic liquids as vaccine adjuvants
- GROMACS + Julia workflows for physicochemical and trajectory analysis

Jan 2021 – Oct 2023

MSc in Physical Chemistry `computational`

University of Campinas (UNICAMP)

- Molecular dynamics of antigen–antibody interactions for medicinal chemistry · thesis (PDF)
- Intrinsically disordered regions and how mutations shift the balance between order and disorder

2016

Summer Intern, Artificial Intelligence `computational`

University of São Paulo (EACH-USP) · São Paulo, Brazil

Jan 2015 – Mar 2017

Scientific Initiation, Molecular Modeling Laboratory `computational`

Federal University of Pará (UFPA) · Belém, Brazil

Mar 2014 – Mar 2018

BSc in Industrial Chemistry `experimental`

Federal University of Pará (UFPA) · Belém, Brazil

Publications & outputs

2024 **Extended Conformational Selection in the Antigen–Antibody Interaction of the PfAMA1 Protein**
P. C. C. da Silva, L. Martínez
The Journal of Physical Chemistry B, 2024

2024 **Intrinsically Disordered Malaria Antigens: An Overview of Structures, Dynamics and Molecular Simulation Opportunities**
P. C. C. da Silva, L. Martínez
Journal of the Brazilian Chemical Society, 35(10), 2024

2023 **Molecular dynamics study of the antigen–antibody interaction mechanism of the Plasmodium falciparum AMA1 protein**
P. C. C. da Silva
MSc thesis, UNICAMP

06 I can solve problems in

🇧🇷 **Português**
native · thinks in it

🇬🇧 **English**
professional · works in it daily

🐍 **Python**
data wrangling & scripts

⚡ **Julia**
fluent · speaks it to GROMACS

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